Classification Models, Evaluation Metrics & Handling Imbalanced Data

Objective

The objective of this task was to build and evaluate classification models using a real-world dataset. The project focused on predicting customer churn, understanding classification metrics, analyzing model performance, and handling imbalanced data for better predictions.

Dataset Used

The **Telco Customer Churn Dataset** was used in this project. It contains customer information such as gender, tenure, monthly charges, total charges, internet service, contract type, payment method, and churn status.

The target variable was **Churn**, where:

- 1 = Customer left the service
- 0 = Customer stayed with the service

Tools and Technologies

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

Methodology

1. Data Preparation

The dataset was loaded successfully and checked for missing values. The TotalCharges column was converted into numeric format, and missing values were filled using the median value. The customerID column was removed because it does not contribute to prediction.

2. Data Encoding

Since the dataset contained categorical values such as Yes/No and service types, categorical columns were converted into numerical format using Label Encoding.

3. Feature Scaling

Feature scaling was applied using StandardScaler to normalize input values and improve model performance.

4. Train-Test Split

The dataset was divided into training and testing sets in an 80:20 ratio. The training set was used for model training, and the testing set was used for final evaluation.

5. Logistic Regression

A Logistic Regression model was trained as the primary classification model. Predictions were made on the test data and evaluated using confusion matrix, precision, recall, F1-score, and accuracy.

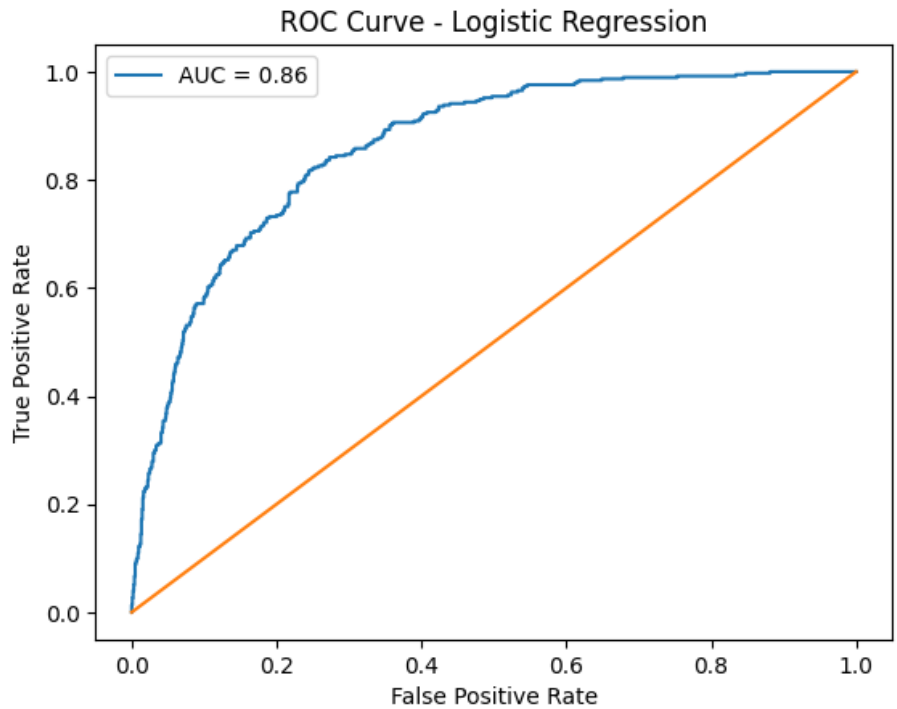
```
Logistic Regression Results
[[933 103]
 [157 216]]
              precision    recall  f1-score   support

           0       0.86      0.90      0.88      1036
           1       0.68      0.58      0.62       373

    accuracy                           0.82      1409
   macro avg       0.77      0.74      0.75      1409
weighted avg       0.81      0.82      0.81      1409
```

6. ROC Curve and AUC

The ROC Curve was plotted to analyze the trade-off between True Positive Rate and False Positive Rate. The AUC score was used to measure the model's classification ability.



7. Handling Imbalanced Data

Customer churn datasets are often imbalanced because the number of customers who stay is higher than those who leave. To address this, Logistic Regression with class_weight = balanced was used.

```
Balanced Logistic Regression
[[754 282]
 [ 61 312]]
              precision    recall  f1-score   support

           0       0.93      0.73      0.81      1036
           1       0.53      0.84      0.65       373

    accuracy                           0.76      1409
   macro avg       0.73      0.78      0.73      1409
weighted avg       0.82      0.76      0.77      1409
```

```
Decision Tree Results
[[881 155]
 [135 238]]
              precision    recall  f1-score   support

           0       0.87      0.85      0.86      1036
           1       0.61      0.64      0.62       373

    accuracy                           0.79      1409
   macro avg       0.74      0.74      0.74      1409
weighted avg       0.80      0.79      0.80      1409
```

8. Model Comparison

A Decision Tree Classifier was also trained and compared with Logistic Regression to evaluate which model performed better.

Results

The Logistic Regression model performed well in predicting customer churn and provided balanced classification results.
The balanced Logistic Regression model improved minority class prediction.
The Decision Tree model was also effective, but Logistic Regression gave more stable and interpretable results.

Graph Analysis

The ROC Curve showed the classification performance of the Logistic Regression model. A curve closer to the top-left corner indicates better prediction capability. The AUC score confirmed good model performance.

Conclusion

This task helped in understanding the complete workflow of classification problems in machine learning. It showed that accuracy alone is not enough for classification tasks, especially when the dataset is imbalanced. Metrics such as precision, recall, F1-score, and ROC-AUC are important for proper evaluation. Handling class imbalance improved prediction quality, and Logistic Regression was selected as the most suitable model for customer churn prediction.